

Quantum Entropy Increase Theorem

Alethfeld Proof Graph

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Abstract

This document presents a formal proof of the Quantum Entropy Increase Theorem, which establishes that the spectral entropy of a diagonal observable L_f associated with a Boolean function $f : \{0, 1\}^n \rightarrow \{+1, -1\}$ increases by exactly the classical influence when the $T^{\otimes n} H^{\otimes n}$ transformation is applied. The proof proceeds through six supporting lemmas establishing properties of Pauli operators under Hadamard and T gate conjugation, culminating in the main three-part theorem.

1 Axioms and Definitions

Definition 1.1 (Boolean Function). A Boolean function is a mapping $f : \{0, 1\}^n \rightarrow \{+1, -1\}$.

Definition 1.2 (Fourier Expansion). Every Boolean function f has a unique Fourier expansion

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S(x)$$

where $\chi_S(x) = (-1)^{\sum_{i \in S} x_i}$ is the parity function and $\hat{f}(S) = \mathbb{E}_x[f(x) \chi_S(x)]$ are the Fourier coefficients.

Definition 1.3 (Pauli Operators). The single-qubit Pauli matrices are:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

For $S \subseteq [n]$, define $Z_S = \bigotimes_{i=1}^n P_i$ where $P_i = Z$ if $i \in S$ and $P_i = I$ otherwise.

Definition 1.4 (Hadamard Gate). The Hadamard gate is $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Note that $H^\dagger = H$ and $H^2 = I$.

Definition 1.5 (T Gate). The T gate is $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} = \text{diag}(1, \omega)$ where $\omega = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$.

Definition 1.6 (Spectral Entropy). For a Hermitian operator A with Pauli expansion $A = \sum_P \hat{a}(P)P$, the spectral entropy is

$$H(A) = - \sum_{P \in \mathcal{P}_n} \pi_A(P) \log_2 \pi_A(P)$$

where $\pi_A(P) = |\hat{a}(P)|^2 / \|A\|_2^2$ is the Pauli spectral distribution.

Definition 1.7 (Quantum Influence). The quantum influence of A is

$$\text{Inf}(A) = \sum_{P \in \mathcal{P}_n} \text{wt}(P) \cdot \pi_A(P) = \mathbb{E}_{P \sim \pi_A}[\text{wt}(P)]$$

where $\text{wt}(P) = |\{i : P_i \neq I\}|$ is the Pauli weight.

2 Supporting Lemmas

2.1 Lemma 1: Diagonal Pauli Expansion

Lemma 2.1 (Diagonal Pauli Expansion). For $f : \{0, 1\}^n \rightarrow \{+1, -1\}$, the diagonal observable $L_f = \sum_x f(x)|x\rangle\langle x|$ has Pauli expansion

$$L_f = \sum_{S \subseteq [n]} \hat{f}(S) Z_S$$

where $\hat{f}(S)$ are the Fourier coefficients of f .

Proof. (1)1. Let $f : \{0, 1\}^n \rightarrow \{+1, -1\}$ be a Boolean function with Fourier expansion $f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S(x)$ where $\chi_S(x) = (-1)^{\sum_{i \in S} x_i}$. (Assumption)

(1)2. Define $L_f = \sum_{x \in \{0, 1\}^n} f(x)|x\rangle\langle x|$ as the diagonal observable associated with f . (Definition)

(1)3. Define $Z_S = \bigotimes_{i=1}^n P_i$ where $P_i = Z$ if $i \in S$ and $P_i = I$ otherwise. This is a diagonal Pauli operator with eigenvalues ± 1 . (Definition)

(1)4. For all $x \in \{0, 1\}^n$ and $S \subseteq [n]$: $\langle x | Z_S | x \rangle = (-1)^{\sum_{i \in S} x_i} = \chi_S(x)$. (Algebraic computation)

(2)4.1. Since $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, we have $\langle x_i | Z | x_i \rangle = (-1)^{x_i}$

(2)4.2. For $i \in S$: $\langle x_i | Z | x_i \rangle = (-1)^{x_i}$; for $i \notin S$: $\langle x_i | I | x_i \rangle = 1$

(2)4.3. By tensor product: $\langle x | Z_S | x \rangle = \prod_{i \in S} (-1)^{x_i} = (-1)^{\sum_{i \in S} x_i} = \chi_S(x)$

(1)5. For $\sum_S \hat{f}(S) Z_S$: $\langle x | \sum_S \hat{f}(S) Z_S | x \rangle = \sum_S \hat{f}(S) \chi_S(x) = f(x)$. (Fourier expansion)

(1)6. Since Z_S is diagonal and L_f is diagonal, the off-diagonal elements match (all zero).

(1)7. Therefore $L_f = \sum_{S \subseteq [n]} \hat{f}(S) Z_S$. **QED.**

[Lean: DiagonalObs.lean:93]

□

2.2 Lemma 2: Hadamard Conjugation

Lemma 2.2 (Hadamard Conjugation). *Under conjugation by the Hadamard gate H :*

1. $HXH^\dagger = Z$
2. $HYH^\dagger = -Y$
3. $HZH^\dagger = X$
4. $HIH^\dagger = I$

Consequently, for any $S \subseteq [n]$: $H^{\otimes n} Z_S (H^{\otimes n})^\dagger = X_S$.

Proof. (1)1. The Hadamard gate is $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ with $H^\dagger = H$. (Definition)

(1)2. $HXH^\dagger = Z$: Direct computation shows $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$

(1)3. $HYH^\dagger = -Y$: Direct computation gives the result

(1)4. $HZH^\dagger = X$: Direct computation shows $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X$

(1)5. $HIH^\dagger = I$: Since $H^2 = I$.

(1)6. For $Z_S = \bigotimes_{i \in S} Z_i \otimes \bigotimes_{i \notin S} I_i$, using $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$:

$$H^{\otimes n} Z_S (H^{\otimes n})^\dagger = \bigotimes_{i \in S} (H Z_i H^\dagger) \otimes \bigotimes_{i \notin S} (H I_i H^\dagger) = \bigotimes_{i \in S} X_i \otimes \bigotimes_{i \notin S} I_i = X_S$$

QED.

[Lean: [Gates.lean:57-75](#)]

□

2.3 Lemma 3: T Gate Conjugation

Lemma 2.3 (T Gate Conjugation). *Under conjugation by the T gate $T = \text{diag}(1, e^{i\pi/4})$:*

1. $TIT^\dagger = I$
2. $TZT^\dagger = Z$
3. $TXT^\dagger = \frac{1}{\sqrt{2}}(X + Y)$
4. $TYT^\dagger = \frac{1}{\sqrt{2}}(Y - X)$

Proof. (1)1. Define $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} = \text{diag}(1, \omega)$ where $\omega = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$ and $T^\dagger = \text{diag}(1, \omega^*) = \text{diag}(1, -i\omega)$ where $\omega^* = e^{-i\pi/4} = \frac{1-i}{\sqrt{2}}$. (Definition)

(1)2. $TIT^\dagger = I$: Since $T \cdot I \cdot T^\dagger = TT^\dagger = I$ (T is unitary).

(1)3. $TZT^\dagger = Z$: Both T and Z are diagonal, so $\text{diag}(1, \omega) \cdot \text{diag}(1, -1) \cdot \text{diag}(1, \omega^*) = \text{diag}(1, -\omega\omega^*) = \text{diag}(1, -1) = Z$

(1)4. $TXT^\dagger = \frac{1}{\sqrt{2}}(X + Y)$:

$$\begin{pmatrix} 1 & 0 \\ 0 & \omega \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \omega^* \end{pmatrix} = \begin{pmatrix} 0 & \omega^* \\ \omega & 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1-i \\ 1+i & 0 \end{pmatrix} = \frac{1}{\sqrt{2}}(X + Y)$$

⟨1⟩5. $TYT^\dagger = \frac{1}{\sqrt{2}}(Y - X)$:

$$\begin{pmatrix} 1 & 0 \\ 0 & \omega \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \omega^* \end{pmatrix} = \begin{pmatrix} 0 & -i\omega^* \\ i\omega & 0 \end{pmatrix} = \frac{1}{\sqrt{2}}(Y - X)$$

where $i\omega = e^{i3\pi/4} = \frac{-1+i}{\sqrt{2}}$ and $-i\omega^* = e^{-i3\pi/4} = \frac{-1-i}{\sqrt{2}}$. **QED.**

[Lean: [Gates.lean:168-246](#)]

□

2.4 Lemma 4: T Expansion

Lemma 2.4 (T Expansion of X-type Paulis). *Let $S \subseteq [n]$ and $X_S = \bigotimes_{i \in S} X_i \otimes \bigotimes_{i \notin S} I_i$. Then:*

$$T^{\otimes n} X_S (T^{\otimes n})^\dagger = \frac{1}{2^{|S|/2}} \sum_{R \subseteq S} \omega_R \cdot X_{S \setminus R} Y_R I_{S^c}$$

where all $2^{|S|}$ resulting Paulis have weight $|S|$ and coefficients of equal magnitude $2^{-|S|/2}$.

Proof.⟨1⟩1. By Lemma 2.3: $TXT^\dagger = \frac{1}{\sqrt{2}}(X + Y)$ and $TIT^\dagger = I$.

⟨1⟩2. For X_S , using tensor product properties:

$$T^{\otimes n} X_S (T^{\otimes n})^\dagger = \bigotimes_{i \in S} \frac{1}{\sqrt{2}}(X + Y) \otimes \bigotimes_{i \notin S} I$$

⟨1⟩3. Expanding the tensor product over $i \in S$:

$$\bigotimes_{i \in S} \frac{1}{\sqrt{2}}(X_i + Y_i) = \frac{1}{2^{|S|/2}} \sum_{R \subseteq S} \prod_{i \in S \setminus R} X_i \prod_{j \in R} Y_j$$

where each term corresponds to choosing subset $R \subseteq S$ for positions where Y is chosen.

⟨1⟩4. Each term $\prod_{i \in S \setminus R} X_i \prod_{j \in R} Y_j$ has weight exactly $|S|$: there are $|S \setminus R| + |R| = |S|$ non-identity factors.

⟨1⟩5. The number of terms is $2^{|S|}$ (one for each $R \subseteq S$), each with coefficient magnitude $2^{-|S|/2}$. The squared magnitudes are $2^{-|S|}$, summing to 1. **QED.**

[Lean: [TExpansion.lean:35-52](#)]

□

2.5 Lemma 5: Weight Preservation

Lemma 2.5 (Weight Preservation under Product Unitaries). *Let A be a Hermitian operator and $U = U_1 \otimes \cdots \otimes U_n$ be a product of single-qubit unitaries. Then $\text{Inf}(UAU^\dagger) = \text{Inf}(A)$.*

Proof.⟨1⟩1. Recall $\text{Inf}(A) = \sum_{P \in \mathcal{P}_n} \text{wt}(P) \cdot \pi_A(P)$ where $\text{wt}(P) = |\{i : P_i \neq I\}|$. (Definition)

⟨1⟩2. For any single-qubit unitary V , conjugation maps the Pauli group to itself (up to phases): $VIV^\dagger = I$ and $VPV^\dagger \in \{\pm X, \pm Y, \pm Z, \pm iX, \dots\}$ for $P \in \{X, Y, Z\}$. In particular, $\text{wt}(VPV^\dagger) = \text{wt}(P)$.

⟨1⟩3. For product unitary $U = U_1 \otimes \cdots \otimes U_n$ and Pauli string $P = P_1 \otimes \cdots \otimes P_n$:

$$UPU^\dagger = (U_1 P_1 U_1^\dagger) \otimes \cdots \otimes (U_n P_n U_n^\dagger)$$

Since each $U_i P_i U_i^\dagger$ has the same weight contribution as P_i , we have $\text{wt}(UPU^\dagger) = \text{wt}(P)$.

⟨1⟩4. Under conjugation, the Pauli expansion transforms bijectively (up to phases), preserving $\|A\|_2^2$. The spectral distribution satisfies $\pi_{UAU^\dagger}(Q) = \pi_A(U^\dagger QU)$.

⟨1⟩5. Therefore:

$$\text{Inf}(UAU^\dagger) = \sum_Q \text{wt}(Q) \cdot \pi_{UAU^\dagger}(Q) = \sum_P \text{wt}(UPU^\dagger) \cdot \pi_A(P) = \sum_P \text{wt}(P) \cdot \pi_A(P) = \text{Inf}(A)$$

QED.

[Lean: TExpansion.lean:85]

□

2.6 Lemma 6: Entropy of Uniform Splitting

Lemma 2.6 (Entropy of Uniform Splitting). *Let π be a probability distribution on finite set Ω with entropy $H(\pi)$. For each $\omega \in \Omega$, let $k_\omega \geq 1$. Define $\pi'(\omega, j) = \frac{\pi(\omega)}{k_\omega}$ on $\Omega' = \bigsqcup_\omega \{(\omega, j) : j \in [k_\omega]\}$. Then:*

$$H(\pi') = H(\pi) + \sum_{\omega \in \Omega} \pi(\omega) \log_2 k_\omega$$

Proof.⟨1⟩1. Let π be a probability distribution with Shannon entropy $H(\pi) = -\sum_\omega \pi(\omega) \log_2 \pi(\omega)$.

⟨1⟩2. Define $\Omega' = \bigsqcup_{\omega \in \Omega} \{(\omega, j) : j \in [k_\omega]\}$ and $\pi'(\omega, j) = \frac{\pi(\omega)}{k_\omega}$.

⟨1⟩3. Verify π' is a probability distribution: $\sum_{(\omega, j) \in \Omega'} \pi'(\omega, j) = \sum_\omega \sum_{j=1}^{k_\omega} \frac{\pi(\omega)}{k_\omega} = \sum_\omega \pi(\omega) = 1$.

⟨1⟩4. Compute the entropy:

$$H(\pi') = - \sum_{(\omega, j) \in \Omega'} \frac{\pi(\omega)}{k_\omega} \log_2 \frac{\pi(\omega)}{k_\omega}$$

⟨1⟩5. Using $\log_2 \frac{\pi(\omega)}{k_\omega} = \log_2 \pi(\omega) - \log_2 k_\omega$:

$$H(\pi') = - \sum_\omega \sum_{j=1}^{k_\omega} \frac{\pi(\omega)}{k_\omega} \log_2 \pi(\omega) + \sum_\omega \sum_{j=1}^{k_\omega} \frac{\pi(\omega)}{k_\omega} \log_2 k_\omega$$

⟨1⟩6. First term: $\sum_{j=1}^{k_\omega} \frac{\pi(\omega)}{k_\omega} = \pi(\omega)$, so $-\sum_\omega \sum_j \frac{\pi(\omega)}{k_\omega} \log_2 \pi(\omega) = H(\pi)$.

⟨1⟩7. Second term: $\sum_{j=1}^{k_\omega} \frac{\pi(\omega)}{k_\omega} \log_2 k_\omega = \pi(\omega) \log_2 k_\omega$, so $\sum_\omega \pi(\omega) \log_2 k_\omega = \mathbb{E}_\pi[\log_2 k_\omega]$.

⟨1⟩8. Combining: $H(\pi') = H(\pi) + \sum_{\omega \in \Omega} \pi(\omega) \log_2 k_\omega$. **QED.**

[Lean: TExpansion.lean:157]

□

3 Main Result

Theorem 3.1 (Quantum Entropy Increase Theorem). *Let $f : \{0, 1\}^n \rightarrow \{+1, -1\}$ be any Boolean function. Let L_f be its diagonal observable and $\tilde{L}_f = \mathcal{U}(L_f) = T^{\otimes n} H^{\otimes n} L_f (H^{\otimes n})^\dagger (T^{\otimes n})^\dagger$. Then:*

$$(i) \quad H(\tilde{L}_f) = H(L_f) + \text{Inf}(L_f) = H(f) + \text{Inf}(f)$$

$$(ii) \quad \text{Inf}(\tilde{L}_f) = \text{Inf}(L_f) = \text{Inf}(f)$$

$$(iii) \quad \frac{H(\tilde{L}_f)}{\text{Inf}(\tilde{L}_f)} = \frac{H(f)}{\text{Inf}(f)} + 1$$

Proof. $\langle 1 \rangle 1$. Let $f : \{0, 1\}^n \rightarrow \{+1, -1\}$ be a Boolean function. By Lemma 2.1, $L_f = \sum_{S \subseteq [n]} \hat{f}(S) Z_S$ is the Pauli expansion of the diagonal observable.

Part (ii): Influence invariance

$\langle 1 \rangle 2$. The transformation $\mathcal{U} = T^{\otimes n} H^{\otimes n}$ is a product unitary, being a tensor product of single-qubit gates TH on each qubit.

$\langle 1 \rangle 3$. By Lemma 2.5 (Weight Preservation), for any product unitary U : $\text{Inf}(UAU^\dagger) = \text{Inf}(A)$. Applying this twice:

$$\text{Inf}(\tilde{L}_f) = \text{Inf}(T^{\otimes n} H^{\otimes n} L_f (H^{\otimes n})^\dagger (T^{\otimes n})^\dagger) = \text{Inf}(L_f)$$

$\langle 1 \rangle 4$. By Lemma 2.1, the Pauli spectral distribution $\pi_{L_f}(Z_S) = \hat{f}(S)^2$. Thus:

$$\text{Inf}(L_f) = \sum_S |S| \cdot \hat{f}(S)^2 = \text{Inf}(f)$$

$\langle 1 \rangle 5$. **Part (ii) Complete:** $\text{Inf}(\tilde{L}_f) = \text{Inf}(L_f) = \text{Inf}(f)$.

Part (i): Entropy increase

$\langle 1 \rangle 6$. By Lemma 2.1, $L_f = \sum_S \hat{f}(S) Z_S$. By Lemma 2.2:

$$H^{\otimes n} L_f (H^{\otimes n})^\dagger = \sum_S \hat{f}(S) X_S$$

(Hadamard maps $Z_S \rightarrow X_S$).

$\langle 1 \rangle 7$. The Pauli spectral distribution after Hadamard is $\pi(X_S) = \hat{f}(S)^2$ – the distribution is the same as before (just relabeled $Z_S \rightarrow X_S$). Hence $H(H^{\otimes n} L_f (H^{\otimes n})^\dagger) = H(L_f) = H(f)$.

$\langle 1 \rangle 8$. By Lemma 2.4, applying $T^{\otimes n}$ splits each X_S into $2^{|S|}$ Paulis of equal coefficient magnitude $2^{-|S|/2} \cdot |\hat{f}(S)|$. The squared coefficients become $\frac{\hat{f}(S)^2}{2^{|S|}}$ for each of $2^{|S|}$ Paulis.

$\langle 1 \rangle 9$. This is exactly the setup of Lemma 2.6 (Entropy of Uniform Splitting) with $\pi =$ Pauli spectral distribution after $H^{\otimes n}$ (where $\pi(X_S) = \hat{f}(S)^2$) and $k_S = 2^{|S|}$ (splitting factor for each X_S).

$\langle 1 \rangle 10$. By Lemma 2.6:

$$\begin{aligned} H(\tilde{L}_f) &= H(L_f) + \sum_S \pi(X_S) \log_2 k_S \\ &= H(L_f) + \sum_S \hat{f}(S)^2 \log_2(2^{|S|}) \\ &= H(L_f) + \sum_S \hat{f}(S)^2 \cdot |S| \\ &= H(L_f) + \text{Inf}(f) \end{aligned}$$

$\langle 1 \rangle 11$. **Part (i) Complete:** Since $H(L_f) = H(f)$ and $\text{Inf}(L_f) = \text{Inf}(f)$ (by Lemma 2.1), we have:

$$H(\tilde{L}_f) = H(L_f) + \text{Inf}(L_f) = H(f) + \text{Inf}(f)$$

Part (iii): Ratio increase

$\langle 1 \rangle 12$. From Parts (i) and (ii):

$$\frac{H(\tilde{L}_f)}{\text{Inf}(\tilde{L}_f)} = \frac{H(f) + \text{Inf}(f)}{\text{Inf}(f)} = \frac{H(f)}{\text{Inf}(f)} + 1$$

$\langle 1 \rangle 13$. **Theorem 1 Complete.** All three parts established. **QED.**

[Lean: [EntropyIncrease.lean:133](#)]

□

4 Lean 4 Formalization

The proof has been fully formalized in Lean 4 with zero sorries and zero axioms. All lemmas and theorems are completely proved, including the diagonal Pauli expansion, spectral distribution characterization, and the main entropy increase theorem.

4.1 Key Definitions

```
29 /-- Boolean function type:  $\{0,1\}^n \rightarrow \{+1,-1\}$  -/  
30 def BoolFunc (n : N) := (Fin n → Bool) → Z  
31  
32 /-- Classical Fourier entropy -/  
33 noncomputable def fourierEntropy {n : N} (f : BoolFunc n) : R :=  
34   - sum S : Finset (Fin n),  
35     let c := (fourierCoeff f S)^2  
36     if c = 0 then 0 else c * Real.log c / Real.log 2  
37  
38 /-- Quantum spectral entropy -/  
39 noncomputable def spectralEntropy {n : N} (A : QubitMat n) : R :=  
40   - sum P : (Fin n → Fin 4),  
41     let prob := spectralDist A P  
42     if prob = 0 then 0 else prob * Real.log prob / Real.log 2  
43  
44 /-- Quantum influence -/  
45 noncomputable def quantumInfluence {n : N} (A : QubitMat n) : R :=  
46   sum P : (Fin n → Fin 4), pauliWeight P * spectralDist A P
```

4.2 Main Theorem

```
432 /-- Quantum Entropy Increase Theorem (complete statement) -/  
433 theorem quantum_entropy_increase_theorem {n : N} (f : BoolFunc n) :  
434   -- Part (i):  $H(L\tilde{f}) = H(L_f) + \text{Inf}(L_f) = H(f) + \text{Inf}(f)$   
435   spectralEntropy (transformedObs f) = fourierEntropy f +  
436     totalInfluence f /\  
437   -- Part (ii):  $\text{Inf}(L\tilde{f}) = \text{Inf}(L_f) = \text{Inf}(f)$   
438   quantumInfluence (transformedObs f) = totalInfluence f /\  
439   -- Part (iii):  $H(L\tilde{f})/\text{Inf}(L\tilde{f}) = H(f)/\text{Inf}(f) + 1$  (when  $\text{Inf}(f) \neq 0$ )  
440   (totalInfluence f != 0 →  
441     spectralEntropy (transformedObs f) / quantumInfluence (  
442       transformedObs f) =  
443       fourierEntropy f / totalInfluence f + 1) := by  
444   refine <_, _, _>  
445   . rw [entropy_increase, diagonalObs_spectralEntropy_eq,  
446     diagonalObs_quantumInfluence_eq]  
447   . rw [influence_preserved, diagonalObs_quantumInfluence_eq]  
448   . intro hI  
449   exact ratio_increase f hI
```

4.3 Verification Status

Component	Type	Lines	Status
hadamard_conjTranspose	lemma	87–96	Proved
hadamard_conj_X	theorem	110–116	Proved
hadamard_conj_Y	theorem	119–125	Proved
hadamard_conj_Z	theorem	128–134	Proved
tgate_conj_I	theorem	204–209	Proved
tgate_conj_Z	theorem	212–217	Proved
tgate_conj_X	theorem	244–254	Proved
tgate_conj_Y	theorem	257–269	Proved
tExpansion_weight_preserved	theorem	283–297	Proved
tExpansion_card	theorem	300–318	Proved
diagonal_pauli_expansion	theorem	93	Proved
diagonalObs_spectralEntropy_eq	theorem	ZIndexEquiv.lean:78	Proved
diagonalObs_quantumInfluence_eq	theorem	ZIndexEquiv.lean:95	Proved
th_transform_entropy_increase	theorem	81	Proved
th_transform_influence_preserved	theorem	99	Proved
entropy_increase	theorem	406–409	Proved
influence_preserved	theorem	412–414	Proved
ratio_increase	theorem	422–427	Proved
quantum_entropy_increase_theorem	theorem	432–445	Proved

Summary: All 19 theorems/lemmas fully proved, 0 axioms, 0 sorries.